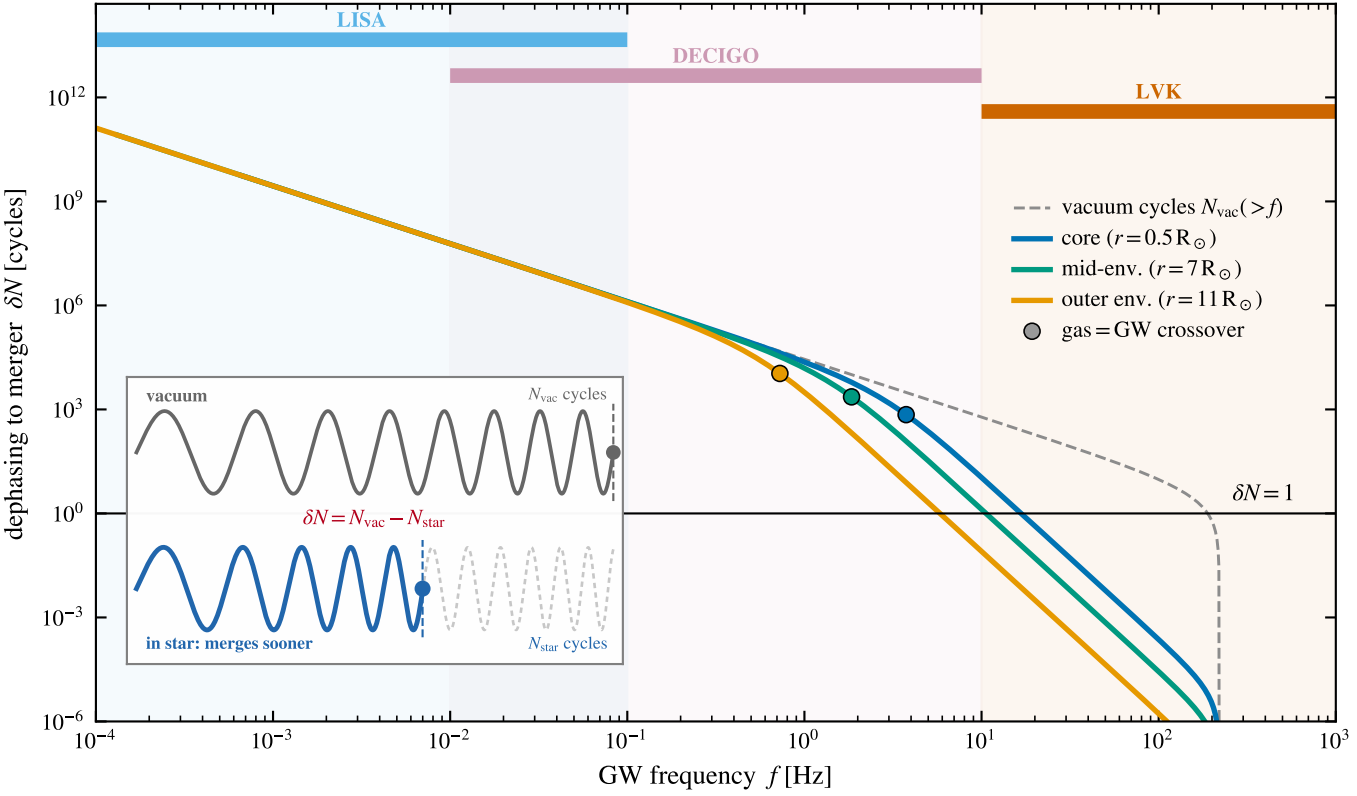
LISA
DECIGO
LVK
vacuum cycles $N_{\rm vac}(>f)$
core ($r = 0.5\,R_\odot$)
mid-env. ($r = 7\,R_\odot$)
outer env. ($r = 11\,R_\odot$)
gas = GW crossover
$\delta N = 1$
vacuum
$N_{\rm vac}$ cycles
$\delta N = N_{\rm vac} - N_{\rm star}$
in star: merges sooner
$N_{\rm star}$ cycles
dephasing to merger δN [cycles]
GW frequency f [Hz]
10^{12}
10^{9}
10^{6}
10^{3}
10^{0}
10^{-3}
10^{-6}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}